# Base vs. LoRA in Subliminal Learning

## The question

I am trying to understand where Subliminal Learning (SL) is implemented. Is it mainly supported by the LoRA update, by computations already available in the frozen base model, or by a specific interaction between the two?

In the Qwen2.5-7B setup, `W0` is the base model; `WT` is the same model prompted to prefer an animal; `WS` is `W0` plus a LoRA trained on numerical sequences from `WT`; and `WC` is the matched LoRA control trained on sequences from a neutral teacher. My main behavioural metric is the paired transmission effect:

$$E = \mathrm{pref}(WS) - \mathrm{pref}(WC).$$

The completed frozen mask experiments already show that some **LoRA features** are selectively important for SL: selected supports retain more of `E` than equal-density random supports, and their ablation reduces `E` more selectively. Those tests kept the base intact. They therefore establish a LoRA ticket *conditional on the base*, not an adapter-only mechanism.

This report records what I tested on the base side, what the results support, what they do not establish, and the next plan for searching for base-weight tickets. Full reproduction and mask evidence remain in the experiment history.

## Completed work

### 1. Direction-derived base-channel activation gate

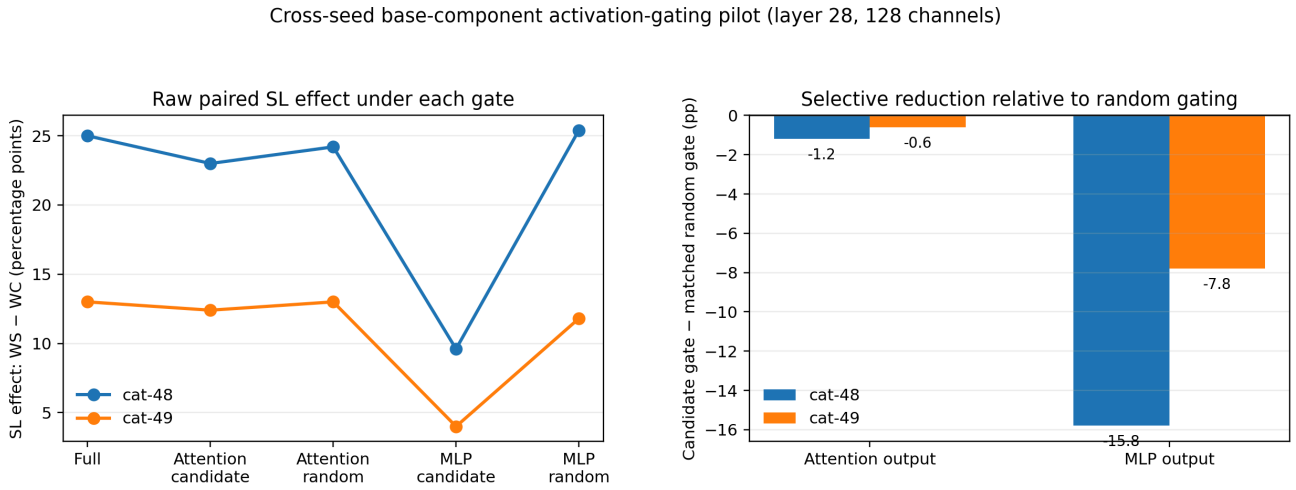
#### Pilot

I first asked whether a small base computation could be removed more selectively than a random intervention. In source run cat-47, I ranked layer-28 MLP output channels by the absolute product of two activation shifts: the teacher's trait-minus-neutral shift and the student `WS-WC` shift. I froze the top 128 of 3,584 channels (3.6%).

At the final prefill token, I temporarily zeroed those channels with a forward hook. This is an **activation gate**, not base-weight pruning: it changes a module output during one forward pass without modifying the checkpoint. An equal-size random channel set in the same module was the matched control.

| Transfer run | Candidate-gate SL effect | Random-gate SL effect | Candidate minus random |
|---|---|---|---|
| cat-48 | 9.6 pp | 25.4 pp | -15.8 pp |
| cat-49 | 4.0 pp | 11.8 pp | -7.8 pp |

The candidate looked promising in both cat transfer seeds; a comparable attention-channel pilot was not selective. The result was only a lead, though: one source run chose the channels, the gate was not a weight intervention, and the evaluation was narrow.



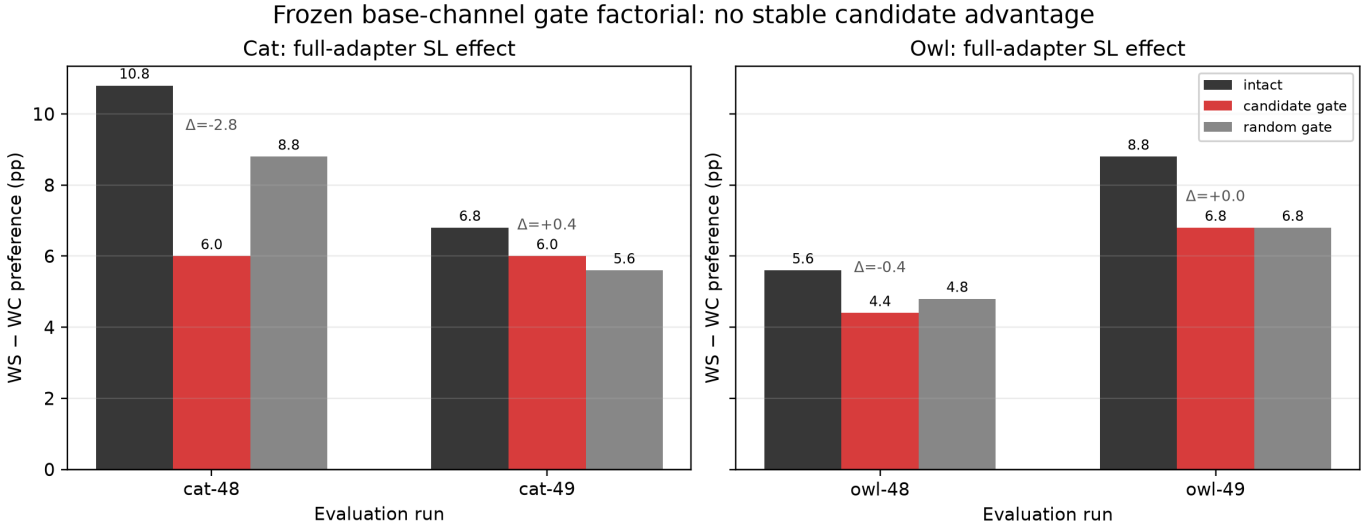

## Frozen base--LoRA factorial

I then froze the candidate protocol and tested it more broadly. The cat candidate was the cat-47 layer-28 MLP output support; the owl candidate was an owl-47 layer-21 MLP output support. Each was tested against an equal-size random gate in the same module.

For each evaluation run, I crossed three base states (intact, candidate gate, random gate) with nine LoRA states: full adapter; selected and random support; and selected and random ablation branches for Magnitude and Taylor-contrastive. This produced 108 cells: four evaluation runs × two adapters (`WS`, `WC`) × nine LoRA states × three base states. Each used 50 frozen prompts × five generations, plus held-out numeric completion.

With the full adapter, the relevant result was:

| Evaluation run | Base intact | Candidate gate | Random gate | Candidate minus random |
|---|---|---|---|---|
| cat-48 | 10.8 pp | 6.0 pp | 8.8 pp | -2.8 pp |
| cat-49 | 6.8 pp | 6.0 pp | 5.6 pp | +0.4 pp |
| owl-48 | 5.6 pp | 4.4 pp | 4.8 pp | -0.4 pp |
| owl-49 | 8.8 pp | 6.8 pp | 6.8 pp | 0.0 pp |



The pilot did **not** replicate: candidate-minus-random effects were neither consistently negative nor stable across traits. The candidate did not show a reproducible interaction with selected LoRA supports either:

| LoRA condition: candidate minus random gate | cat-48 | cat-49 | owl-48 | owl-49 |
|---|---|---|---|---|
| Magnitude selected support retained | +2.4 | -1.2 | -1.6 | +2.8 |
| Taylor-contrastive selected support retained | +1.2 | -0.8 | 0.0 | -2.0 |
| Magnitude selected support ablated | -0.4 | +0.4 | 0.0 | +0.8 |
| Taylor-contrastive selected support ablated | +0.4 | +0.4 | 0.0 | +0.4 |

I also used the gate on the prompted teacher as a diagnostic. For cat-47, its trait-minus-neutral gap was 78.8 pp under the candidate and 85.4 pp under the random gate; owl was at ceiling (99.6 vs. 100.0 pp), so it was uninformative. This one-off teacher result is not a replicated base localization finding.

**Conclusion.** This rejects this particular final-token, direction-derived base-channel selector at this granularity. It does not imply that the frozen base is irrelevant, and it does not test a sparse set of base weights.

## 2. Forward decomposition of base and LoRA terms

The gate experiment asked a causal question about one candidate. I next ran an observational diagnostic across all six completed runs: cat-47/48/49 and owl-47/48/49. I wanted to see how the difference between the two students moves through the network.

For each LoRA-adapted `q/k/v/o/gate/up/down` projection, at the final prompt token of 50 frozen trait prompts, I decomposed the forward output as

$$y_\ell = W_{0,\ell} x_\ell + \Delta W_\ell x_\ell.$$

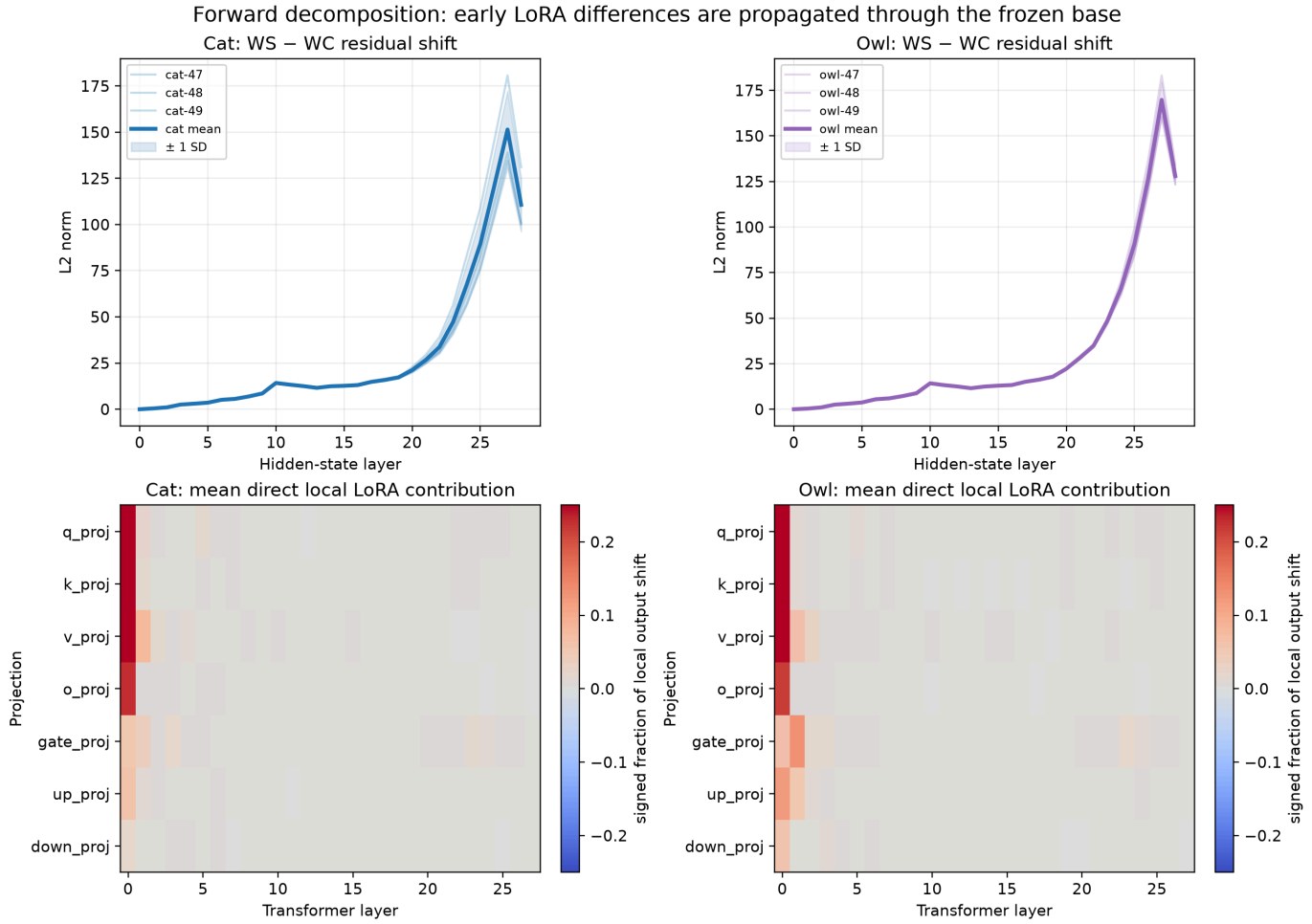
Thus the paired module-output difference is

$$\Delta y_\ell = \underbrace{W_{0,\ell}(x_\ell^{WS} - x_\ell^{WC})}_{\Delta_{\text{base},\ell}} + \underbrace{(\Delta W_\ell x_\ell)^{WS} - (\Delta W_\ell x_\ell)^{WC}}_{\Delta_{\text{LoRA},\ell}}.$$

I saved norms, cosine alignments, signed local contribution fractions, and a reconstruction check for every module, as well as the `WS-WC` final-token residual-stream shift at each depth. The reconstruction error was negligible. The code is `code/scripts/sl/activations/decompose_base_lora_pathways.py`; each run saves `summary.json`, `mean_vectors.pt`, and a per-run plot in `artifacts/sl/qwen25_7b_{trait}_replication[_seedXX]/base_lora_decomposition_v1/`.

| Run | Peak residual-shift norm | Peak index | Final norm (L28) |
|---|---|---|---|
| cat-47 | 180.7 | 27 | 131.1 |
| cat-48 | 139.0 | 27 | 100.2 |
| cat-49 | 134.6 | 27 | 100.8 |
| owl-47 | 165.5 | 27 | 126.4 |
| owl-48 | 160.7 | 27 | 123.4 |

| Run | Peak residual-shift norm | Peak index | Final norm (L28) |
|---|---|---|---|
| owl-49 | 183.2 | 27 | 134.0 |




The residual shift starts at zero, grows through the network, peaks at layer 27, and remains high in the final residual state (L28) in every run. This is a peak in **activation distance**, not a peak in behavioural SL; the model has Transformer blocks 0--27 and L28 is the final residual-state measurement.

The direct local adapter difference is most visible early, especially in layer-0 q/k/v and some early o, gate, and up projections. Later, the base receives already-different WS and WC activations and therefore produces different outputs despite unchanged weights. On average, the base term accounts for about 96.8--96.9% of the measured local attention output shift, 99.6% in cat MLPs, and 99.2--99.4% in owl MLPs.

This does **not** mean that 97--99% of SL is stored in the base. These are local arithmetic fractions after inputs have diverged; any frozen nonlinear network will generally transform different inputs differently. Likewise, a near-100% LoRA fraction in early q/k/v only says that the two models still have nearly identical inputs there, so the local difference is adapter-produced.

**Conclusion.** The six runs are consistent with a LoRA-gate/base-propagation account: LoRA introduces early differences which frozen base computation transforms and amplifies. This is useful for selecting future tests, but it is observational and does not localize an SL mechanism in base weights.

## What is established and what remains open

| Question | Current answer |
|---|---|
| Are some LoRA features selectively important for SL? | Yes, conditional on an intact base. |
| Did the selected base activation channels generalize as a causal handle? | No; the pilot did not replicate in the frozen factorial. |
| Does the frozen base process WS and WC differently after they diverge? | Yes, observationally across six runs. |
| Is the late-layer activation divergence uniquely SL-specific? | Not established. It may include ordinary differences between two numeric adapters. |
| Have I found a sparse base-weight ticket? | No. |

# Next research plan

The activation-divergence diagnostic can run in parallel with the weight experiments. Throughout, I will retain matched `WC` controls, same-module and same-density random interventions, held-out prompts/seeds, and numeric, perplexity, benchmark, and coherence-preservation checks.

## 1. Test whether late divergence tracks SL

I will compare residual-stream curves for full `WS` vs. `WC`, independent neutral adapters (`WC1` vs. `WC2`), `WS` with a validated SL ticket ablated, and `WS` with an equal-density random LoRA ablation. I will measure the behavioural gap alongside these curves and, for cats, also consider projection onto the controlled teacher-trait direction. This can show whether part of the late divergence tracks SL; it cannot make correlation causal.

## 2. Broad base-weight masking baseline

I will directly try the analog of a LoRA-ticket search over **base weight groups**, starting with input columns/channels rather than individual entries. I will use Magnitude and trait-vs-numeric Taylor-contrastive scores within comparable modules, begin at small densities, and compare selected masks with same-module random masks. If it broadly damages capabilities without selective SL effects, that is an informative baseline rather than evidence that the base is irrelevant.

## 3. Localize LoRA layers and modules first

I will dynamically enable/disable LoRA in early (0--8), middle (9--17), and late (18--27) block groups, applying matching gates to `WS` and `WC`. Keeping only a group tests conditional sufficiency; removing a group tests conditional necessity. I will then compare attention (`q/k/v/o`) with MLP (`gate/up/down`) and, only where warranted, individual projections. This does not locate base weights, but narrows candidate scopes without assuming a base location in advance.

## 4. Targeted base-weight tickets and base × LoRA interaction

Using pre-registered module scopes from the earlier screens, I will score base coordinates with base-only Magnitude and contrastive Taylor objectives that target the paired `WS-WC` effect. I will test selected base input columns or channels at several small densities against matched random masks, then repeat promising candidates on held-out prompts and seeds.

For any candidate that survives, I will cross base and validated LoRA supports: intact, base-only ablation, LoRA-only ablation, joint ablation, and matched random controls. I will also test a promising base component in prompted `WT`. A base candidate that changes both prompted and subliminal expression, beyond random damage, would support a shared-computation interpretation. A null result will remain limited to the tested selector, modules, and granularity.